

CHAPTER 4 GAS MIXTURES

To determine the properties of a mixture, we need to know the composition of the mixture as well as the properties of the individual components.

Mass and Mole number of the mixture

Subscripting

- m denotes a gas mixture
- i denotes single component of a mixture
- The mass of a mixture : $m_m = \sum_{i=1}^k m_i$

Number of Moles: $N_m = \sum_{i=1}^k N_i$

- There are two ways to describe the composition of a mixture:
- **Molar analysis**: Specifying the number of moles of each component
- **Gravimetric Analysis**: Specifying the mass of each component

Mass Fraction (mf_i)

$$mf_i = \frac{m_i}{m_m}$$

$$\sum_{i=1}^k mf_i = 1$$

Mole Fraction:

$$y_i = \frac{N_i}{N_m}$$

$$\sum_{i=1}^k y_i = 1$$

The mass of a substance can be expressed in terms of the mole number N and molar mass M of the substance as $m = NM$.

$$M = \frac{m}{N} \quad (\text{kg/kmol})$$

Then the apparent (or average) molar mass and the gas constant of a mixture can be expressed as

$$M_m = \frac{m_m}{N_m} = \frac{\sum m_i}{N_m} = \frac{\sum N_i M_i}{N_m} = \sum_{i=1}^k y_i M_i \quad \text{and} \quad R_m = \frac{R_u}{M_m}$$

Mass and mole fractions of a mixture are related by

$$\text{mf}_i = \frac{m_i}{m_m} = \frac{N_i M_i}{N_m M_m} = y_i \frac{M_i}{M_m}$$

Example 1

The composition of moist air is given on a molar basis to be 78 percent N_2 , 20 percent O_2 and 2 percent water vapour. Determine the mass fractions of the constituents of air.

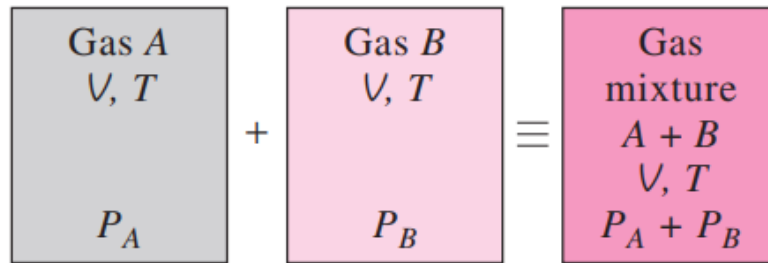
Example 2

Consider a gas mixture that consists of 3 kg of O_2 , 5 kg of N_2 , and 12 kg of CH_4 . Determine (a) the mass fraction of each component, (b) the mole fraction of each component, and (c) the average molar mass and gas constant of the mixture.

P-v-T Behaviour of Gas Mixture

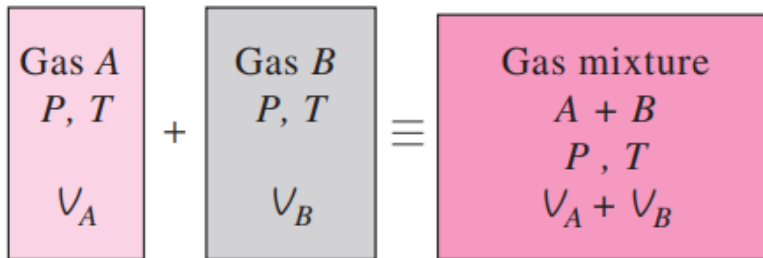
The prediction of the P - v - T behavior of gas mixtures is usually based on two models: *Dalton's law of additive pressures* and *Amagat's law of additive volumes*

Dalton's law of additive pressures: *The pressure of a gas mixture is equal to the sum of the pressures each gas would exert if it existed alone at the mixture temperature and volume.*



$$P_m = \sum_{i=1}^k P_i(T_m, V_m)$$

Amagat's law of additive volumes: *The volume of a gas mixture is equal to the sum of the volumes each gas would occupy if it existed alone at the mixture temperature and pressure.*



$$V_m = \sum_{i=1}^k V_i(T_m, P_m)$$

- In these relations, P_i is called the **component pressure** and V_i is called the **component volume**.
- For ideal gases, P_i and V_i can be related to y_i by using the ideal-gas relation for both the components and the gas mixture: (**$PV = NRT$**)

$$\frac{P_i(T_m, V_m)}{P_m} = \frac{N_i R_u T_m / V_m}{N_m R_u T_m / V_m} = \frac{N_i}{N_m} = y_i$$

$$\frac{V_i(T_m, P_m)}{V_m} = \frac{N_i R_u T_m / P_m}{N_m R_u T_m / P_m} = \frac{N_i}{N_m} = y_i$$

$$\frac{P_i}{P_m} = \frac{V_i}{V_m} = \frac{N_i}{N_m} = y_i$$

Partial pressure : $P_i = P_m y_i$

Partial Volume: $V_i = v_m y_i$

Properties of Gas Mixtures

- A simple way of evaluating the **extensive properties** of a non reacting ideal or real-gas mixture *Just add the contributions of each component of the mixture.*
- Then the total internal energy, enthalpy, and entropy of a gas mixture can be expressed, respectively, as:

$$U_m = \sum_{i=1}^k U_i = \sum_{i=1}^k m_i u_i = \sum_{i=1}^k N_i \bar{u}_i \quad (\text{kJ})$$

$$H_m = \sum_{i=1}^k H_i = \sum_{i=1}^k m_i h_i = \sum_{i=1}^k N_i \bar{h}_i \quad (\text{kJ})$$

$$S_m = \sum_{i=1}^k S_i = \sum_{i=1}^k m_i s_i = \sum_{i=1}^k N_i \bar{s}_i \quad (\text{kJ/K})$$

the changes in internal energy, enthalpy, and entropy of a gas mixture during a process can be expressed, respectively, as

$$\Delta U_m = \sum_{i=1}^k \Delta U_i = \sum_{i=1}^k m_i \Delta u_i = \sum_{i=1}^k N_i \Delta \bar{u}_i \quad (\text{kJ})$$

$$\Delta H_m = \sum_{i=1}^k \Delta H_i = \sum_{i=1}^k m_i \Delta h_i = \sum_{i=1}^k N_i \Delta \bar{h}_i \quad (\text{kJ})$$

$$\Delta S_m = \sum_{i=1}^k \Delta S_i = \sum_{i=1}^k m_i \Delta s_i = \sum_{i=1}^k N_i \Delta \bar{s}_i \quad (\text{kJ/K})$$

The internal energy, enthalpy, and entropy of a mixture *per unit mass* or *per unit mole* of the mixture can be determined by dividing the equations above by the mass or the mole number of the mixture (*mm* or *Nm*)

$$u_m = \sum_{i=1}^k m f_i u_i \quad (\text{kJ/kg}) \quad \text{and} \quad \bar{u}_m = \sum_{i=1}^k y_i \bar{u}_i \quad (\text{kJ/kmol})$$

$$h_m = \sum_{i=1}^k m f_i h_i \quad (\text{kJ/kg}) \quad \text{and} \quad \bar{h}_m = \sum_{i=1}^k y_i \bar{h}_i \quad (\text{kJ/kmol})$$

$$s_m = \sum_{i=1}^k m f_i s_i \quad (\text{kJ/kg} \cdot \text{K}) \quad \text{and} \quad \bar{s}_m = \sum_{i=1}^k y_i \bar{s}_i \quad (\text{kJ/kmol} \cdot \text{K})$$

Similarly, the specific heats of a gas mixture can be expressed as

$$c_{v,m} = \sum_{i=1}^k mf_i c_{v,i} \quad (\text{kJ/kg} \cdot \text{K}) \quad \text{and} \quad \bar{c}_{v,m} = \sum_{i=1}^k y_i \bar{c}_{v,i} \quad (\text{kJ/kmol} \cdot \text{K})$$
$$c_{p,m} = \sum_{i=1}^k mf_i c_{p,i} \quad (\text{kJ/kg} \cdot \text{K}) \quad \text{and} \quad \bar{c}_{p,m} = \sum_{i=1}^k y_i \bar{c}_{p,i} \quad (\text{kJ/kmol} \cdot \text{K})$$

Care should be exercised, however, in evaluating the Δs of the components since the entropy of an ideal gas depends on the pressure or volume of the component as well as on its temperature.

$$\Delta s_i = s_{i,2}^\circ - s_{i,1}^\circ - R_i \ln \frac{P_{i,2}}{P_{i,1}} \cong c_{p,i} \ln \frac{T_{i,2}}{T_{i,1}} - R_i \ln \frac{P_{i,2}}{P_{i,1}}$$

where $P_{i,2} = y_{i,2} P_{m,2}$ and $P_{i,1} = y_{i,1} P_{m,1}$ Are partial pressure.

Example 3. Ethane (C_2H_2) at 20°C and 200 kpa ad methane (CH_4) at 45°C and 200 kpa enter an adiabatic mixing chamber. The mass flow rate of ethane is 9 kg/s, which twice the mass flow rate of methane. Determine (a) the mixture temperature, and (b) the rate of entropy generation during this process, in kw/k. Take $T_o = 25^\circ\text{C}$.